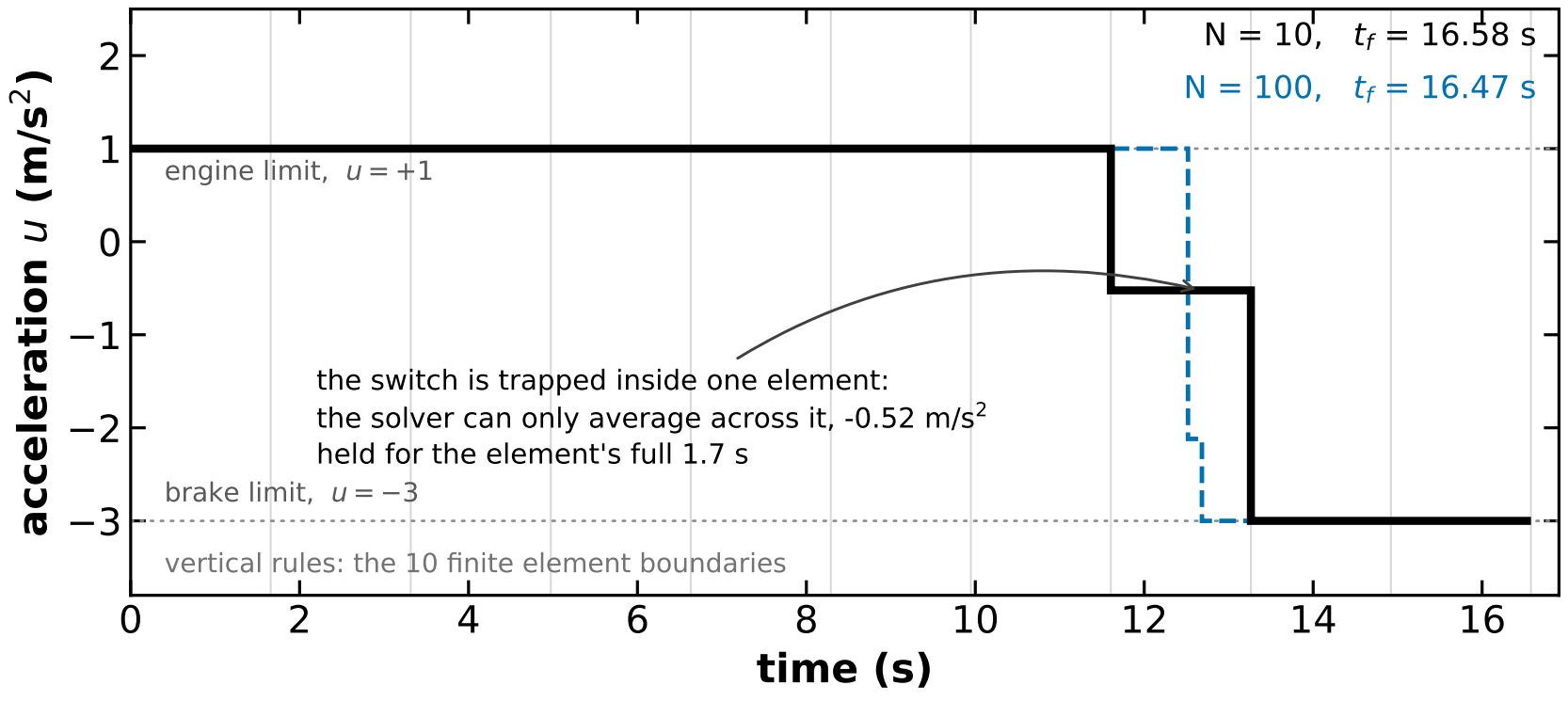
N = 10, $t_f$ = 16.58 s
N = 100, $t_f$ = 16.47 s
engine limit, $u = +1$
the switch is trapped inside one element:
the solver can only average across it, -0.52 m/s$^2$
held for the element's full 1.7 s
brake limit, $u = -3$
vertical rules: the 10 finite element boundaries
acceleration $u$ (m/s$^2$)
time (s)
2
1
0
−1
−2
−3
0
2
4
6
8
10
12
14
16